

Third day of class - 9/2

Fun fact: Consider the equation

$$x^2 - 34y^2 = -1.$$

This equation has

- (a) Solutions modulo n for every $n \geq 1$.
- (b) Rational solutions (like $x = 5/3, y = -1/3$).
- (c) No integer solutions.

Curves of genus 0

An elliptic curve is a degree three equation, $y^2 = x^3 + Ax + B$. Before we get to the study of elliptic curves, I'd like to talk a little bit about simpler cases, show you how to find all solutions to certain equations.

It turns out that *any* cubic equation in two variables can be put into the form of an elliptic curve, $y^2 = x^3 + Ax + B$ with $A, B \in \mathbb{Z}$, provided one can find at least one rational solution to the equation.

What about lower degree equations? Linear equations, those of the form $ax + by + c = 0$ are very easy to solve. If a is not zero, we can solve for x in terms of y , namely $ax = -by - c$ and so $x = -\frac{b}{a}y - \frac{c}{a}$. If a, b and c are rational, it is clear there are infinitely many rational solutions (because we can take y to be any rational value).

Quadratic equations are a little bit more involved.

Example 1: Find all the rational points on $x^2 + y^2 = 1$.

Method:

- 1) First, find one point. We'll start with $x = -1$ and $y = 0$.
- 2) Now, if (x, y) is any solution with $x, y \in \mathbb{Q}$ other than $(-1, 0)$, the line through $(-1, 0)$ and (x, y) has slope $m = \frac{y-0}{x-(-1)} = \frac{y}{x+1}$, which is a rational number.

We will solve for x and y in terms of m . We have $x^2 + y^2 = 1$. Using that $y = m(x+1)$ we get $x^2 + (m(x+1))^2 = 1$. This gives $x^2 + m^2(x^2 + 2x + 1) = 1$ and so $(1+m^2)x^2 + 2m^2x + (m^2-1) = 0$. That looks like a mess!

- 3) Now, there's a trick. The roots of this equation are the x -coordinates of the intersection points of the circle with the line through $(-1, 0)$ with slope m . This means that $(x+1)$ is a factor of the left hand side. That makes it easy to factor. We get

$$(1+m^2)x^2 + 2m^2x + (m^2-1) = (x+1)((1+m^2)x + (m^2-1)) = 0.$$

- 4) Since $x \neq -1$, we get $(1+m^2)x + (m^2-1) = 0$ and so $(1+m^2)x = 1-m^2$ or $x = \frac{1-m^2}{1+m^2}$.

5) Plugging in to $y = m(x + 1)$ we get

$$\begin{aligned} y &= m \cdot \left(\frac{1 - m^2}{1 + m^2} + 1 \right) \\ &= m \cdot \left(\frac{1 - m^2}{1 + m^2} + \frac{1 + m^2}{1 + m^2} \right) \\ &= m \cdot \left(\frac{(1 - m^2) + (1 + m^2)}{1 + m^2} \right) \\ &= m \cdot \frac{2}{1 + m^2} = \frac{2m}{1 + m^2}. \end{aligned}$$

We found the point $\left(\frac{1-m^2}{1+m^2}, \frac{2m}{1+m^2} \right)$. Hence, every rational solution other than $(-1, 0)$ has the form above. (You can get the solution $(-1, 0)$ by taking

$$\lim_{m \rightarrow \infty} \frac{1 - m^2}{1 + m^2} = -1, \quad \lim_{m \rightarrow \infty} \frac{2m}{1 + m^2} = 0.$$

)

Note: In the formula we found, we were able to solve for x and y in terms of m . This shows that the x and y we got are rational points.

There's a more conceptual way to see this. Because we're taking the intersection of a line and a circle, the x -coordinates of those intersection points will satisfy a degree 2 equation. If this equation has the form

$$x^2 + ax + b = (x - r_1)(x - r_2) = 0$$

where $a, b \in \mathbb{Q}$, then since we know from the beginning that one of the roots is rational (namely $r_1 = -1$), then since $a = -r_1 - r_2$ is rational, we have that $r_2 = -r_1 - a$ must be rational as well. This sort of reasoning will be important when we talk about elliptic curves.

Application - Pythagorean triples

The solutions to $X^2 + Y^2 = Z^2$ are called Pythagorean triples. These show up in a number of interesting places in mathematics. There is a Babylonian clay tablet (Plimpton 322, from around 1800 BC). (See the blog post titled "Don't fall for Babylonian Trigonometry Hype" by Evelyn Lamb at <https://www.scientificamerican.com/blog/roots-of-unity/dont-fall-for-babylonian-trigonometry-hype/>.)

A solution to this equation is primitive if $\gcd(X, Y, Z) = 1$. We have that $(X/Z, Y/Z)$ is then a rational point on $x^2 + y^2 = 1$ and so $X/Z = \frac{1-m^2}{1+m^2}$ and $Y/Z = \frac{2m}{1+m^2}$. Setting $m = a/b$, we get $X = a^2 - b^2$, $Y = 2ab$ and $Z = a^2 + b^2$. This is all primitive solutions. (Well, in order to get a primitive solution one of a or b must be odd, and the other must be even.)

Another example

The same procedure that I described above works for any degree two equation with rational coefficients, provided you can find a solution.

However, solutions do not always exist!

Consider $x^2 + y^2 = 3$. Suppose that $x = X/Z$ where $\gcd(X, Z) = 1$. Then, $(X/Z)^2 + y^2 = 3$ and so $X^2 + y^2 Z^2 = 3Z^2$. The first term on the left hand side and the right hand side are integers and so $y^2 Z^2$ is an integer too. Let $y^2 Z^2 = Y^2$. Then, we have $X^2 + Y^2 = 3Z^2$.

Now, we look at the equation modulo 3. We have $X^2 + Y^2 \equiv 0 \pmod{3}$. If $X \equiv 0 \pmod{3}$, then $X^2 \equiv 0 \pmod{3}$. If $X \equiv 1 \pmod{3}$, then $X^2 \equiv 1 \pmod{3}$ and if $X \equiv 2 \pmod{3}$, then $X^2 \equiv 4 \equiv 1 \pmod{3}$. The only way we can have $X^2 + Y^2 \equiv 0 \pmod{3}$ then is if $X \equiv 0 \pmod{3}$ and $Y \equiv 0 \pmod{3}$. This makes $X^2 + Y^2$ a multiple of 9. Since $X^2 + Y^2 = 3Z^2$, this forces Z^2 and hence Z to be a multiple of 3. This contradicts that $\gcd(X, Z) = 1$.

Thus, there are no solutions!

Examples for you folks to try

Parametrize the solutions to $x^2 + y^2 = 2$. This means find a pair of functions $x(m)$ and $y(m)$ so that every rational point (with one exception) is of the form $x = x(m)$, $y = y(m)$. There shouldn't be any square roots in your $x(m)$ and $y(m)$ functions.

Give this a try. If you get stuck, let me know.

For homework:

For each of the following equations, either find a rational point on the curve, or prove that none exists.

- (a) $x^2 + y^2 = 6$.
- (b) $3x^2 + 5y^2 = 4$.
- (c) $3x^2 + 6y^2 = 4$.

General picture

Suppose that C is a curve defined by the equation $C : f(x, y) = 0$. The genus of a curve C is an important invariant of the curve, and it is a non-negative integer.

If you look at the *complex points* on the curve, the solutions where $x = a + bi$ and $y = c + di$ where $a, b, c, d \in \mathbb{R}$, that set of points will either look like a sphere, or a donut with g holes in it. If the complex points look like a sphere, the genus of C is zero. If the complex points look like a donut with g holes in it, the genus is equal to g .

Every conic has genus 0. (In fact, every genus 0 curve is equivalent to a conic by some change of variables.)

If a curve is defined by an equation $f(x, y) = 0$, we say that a point $P = (x_0, y_0)$ on $f(x, y)$ is *singular* if

$$f(x_0, y_0) = 0 \quad \text{and} \quad \frac{\partial f}{\partial x}(x_0, y_0) = 0 \quad \text{and} \quad \frac{\partial f}{\partial y}(x_0, y_0) = 0.$$

(This is really saying that there is no well-defined tangent line to the curve at (x_0, y_0) .) We like curves that are non-singular.

If C is a curve without any singular points defined by $f(x, y)$ of degree d , the genus of C is $d(d-1)/2$.

Theorem. *If C is a curve of genus zero, and C has a rational point that is not singular, then there are infinitely many rational points on C , and they can be parametrized.*

Note: In the event that C is written in the form

$$ax^2 + bxy + cy^2 + dx + ey + f = 0$$

then either there is a rational point on C , or there is a way to prove that C has no rational point using modular arithmetic. (This is called the Hasse-Minkowski theorem.)

Theorem. *If C is a curve of genus one, and C has a rational point on it that is not singular, then C can be put in the form of an elliptic curve $y^2 = x^3 + ax + b$. There can be either finitely many or infinitely many rational points.*

Theorem (Faltings, 1985). *If C is a curve of genus > 1 , then C only has finitely many rational points on it.*

Examples:

1. Any conic has genus zero.
2. Any non-singular cubic curve has genus one. (If a cubic curve is singular it has genus zero.)
3. Are there infinitely many rational numbers $a, b \in \mathbb{Q}$ for which

$$\frac{256(a^2 - 3)^3}{a^2 - 4} = \frac{27(b + 1)(b + 9)^3}{b^3}?$$

This is an odd question to ask, it is connected with the study of elliptic curves. We are asking about the curve

$$256(a^2 - 3)^3 b^3 = 27(a^2 - 4)(b + 1)(b + 9)^3.$$

It turns out that this curve has genus zero. (It has many singular points.) We can parametrize it with $a = \frac{-t^4 + 6t^2 + 3}{4t}$ and $b = \frac{27(t^2 - 1)}{t^6 - 9t^4}$. This shows that any rational number of the form

$$\frac{(t^2 - 3)^3(t^6 - 9t^4 + 3t^2 - 3)^3}{(t^2 - 9)(t^2 - 1)^3 t^4}$$

is of the form $256(a^2 - 3)^3/(a^2 - 4)$ and also of the form $27(b + 1)(b + 9)^3/b^3$.

4. The only rational point on $x^3 + y^3 = 2$ is the point $(1, 1)$. This curve has genus one, and a change of variables takes this curve to the elliptic curve $y^2 = x^3 - 27$.

There are infinitely many rational points on $x^3 + y^3 = 6$. For example,

$$(x, y) = \left(\frac{17}{21}, \frac{37}{21} \right), \left(\frac{2237723}{960540}, -\frac{1805723}{960540} \right).$$

5. The curve $y^2 = 2x^5 - 2$ has genus 2. The only rational points on this curve are $(1, 0)$ and $(3, \pm 22)$. Faltings's theorem does not have a constructive proof, and it is a bit tricky to actually prove these are the only rational points. This problem is a big topic in number theory - how do you provably find all the rational points on some specific curve of genus $g > 1$?